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(54) **METHOD AND APPARATUS FOR INDEXING AND BATCHING SHAFTS**

(71) Applicants: **Mason Springer**, Rushville, OH (US); **Garland Naistetler**, Logan, OH (US)

(72) Inventors: **Mason Springer**, Rushville, OH (US); **Garland Naistetler**, Logan, OH (US)

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(57) **ABSTRACT**

A method and apparatus for indexing and batching shafts. In some embodiments the apparatus for indexing and batching shafts includes a body having upper and lower portions as well as inner and outer portions, a mechanism for applying force to a shaft longitudinally, and the apparatus is configured so the applied force causes the shaft to deflect along the first dynamic bend. The apparatus is configured to attach and detach a single or plurality of gauges.

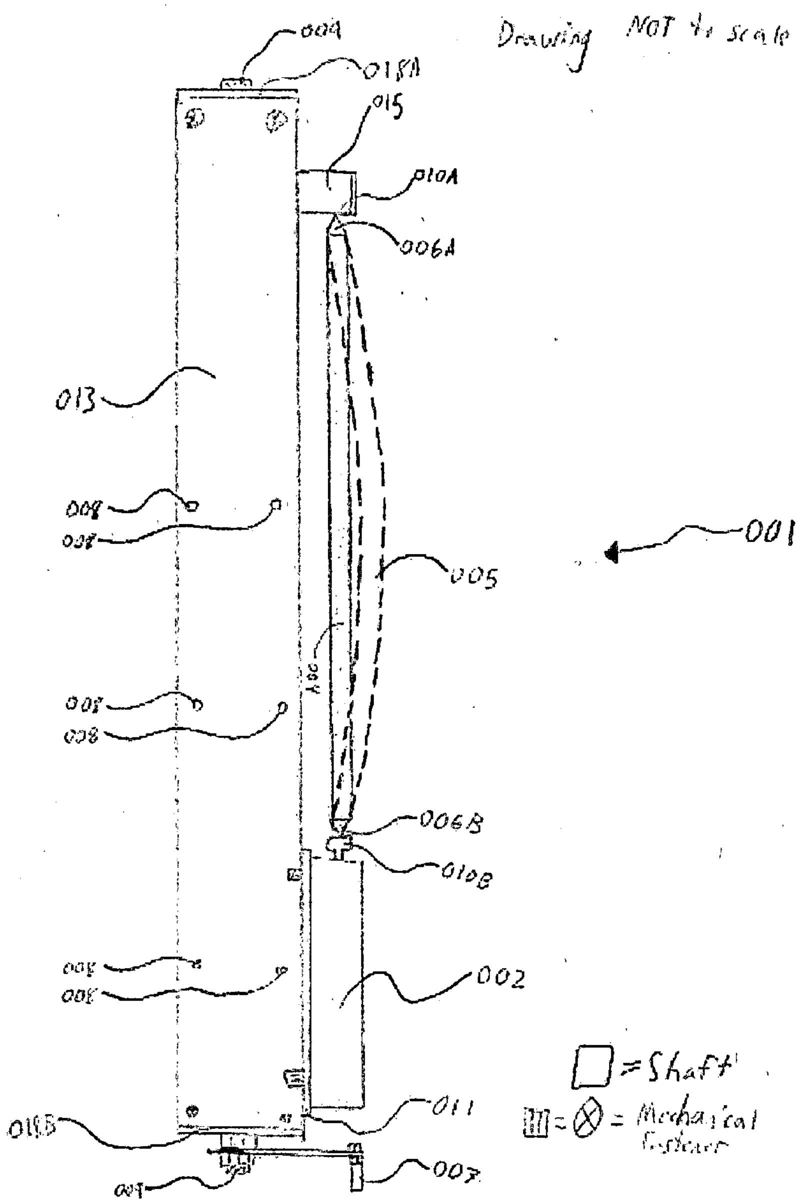


FIG. 1

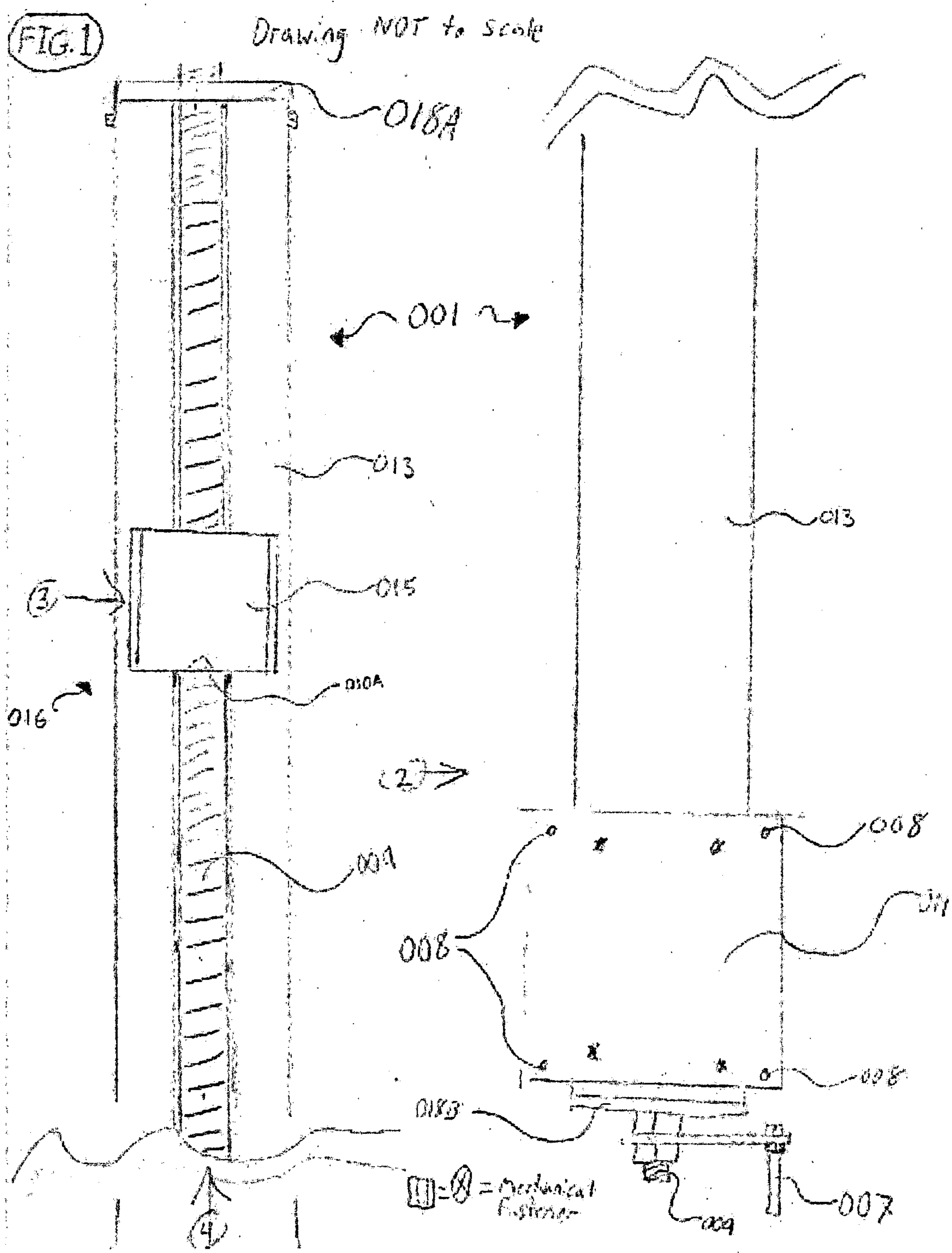

Drawing NOT to scale
018A
001
013
013
3
015
010A
016
2
008
008
014
008
008
018B
009
007
4
= Mechanical Fastener

FIG.2

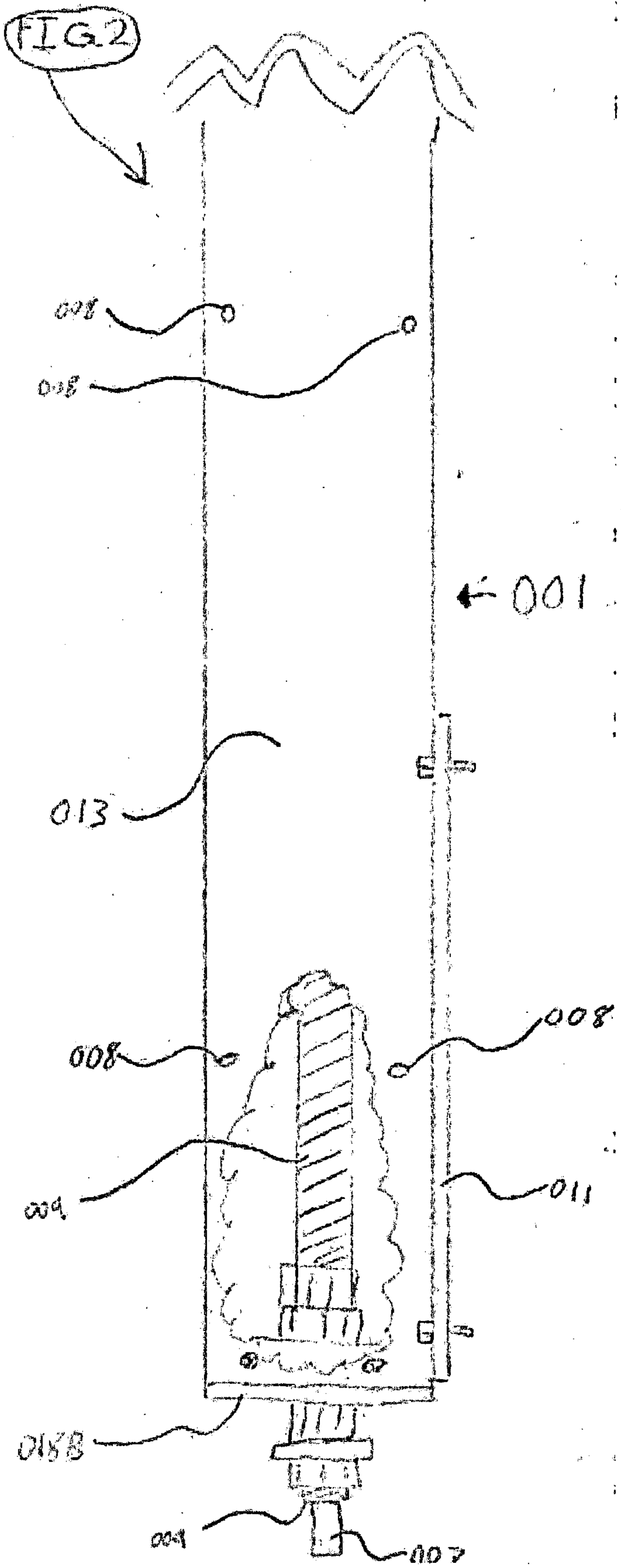
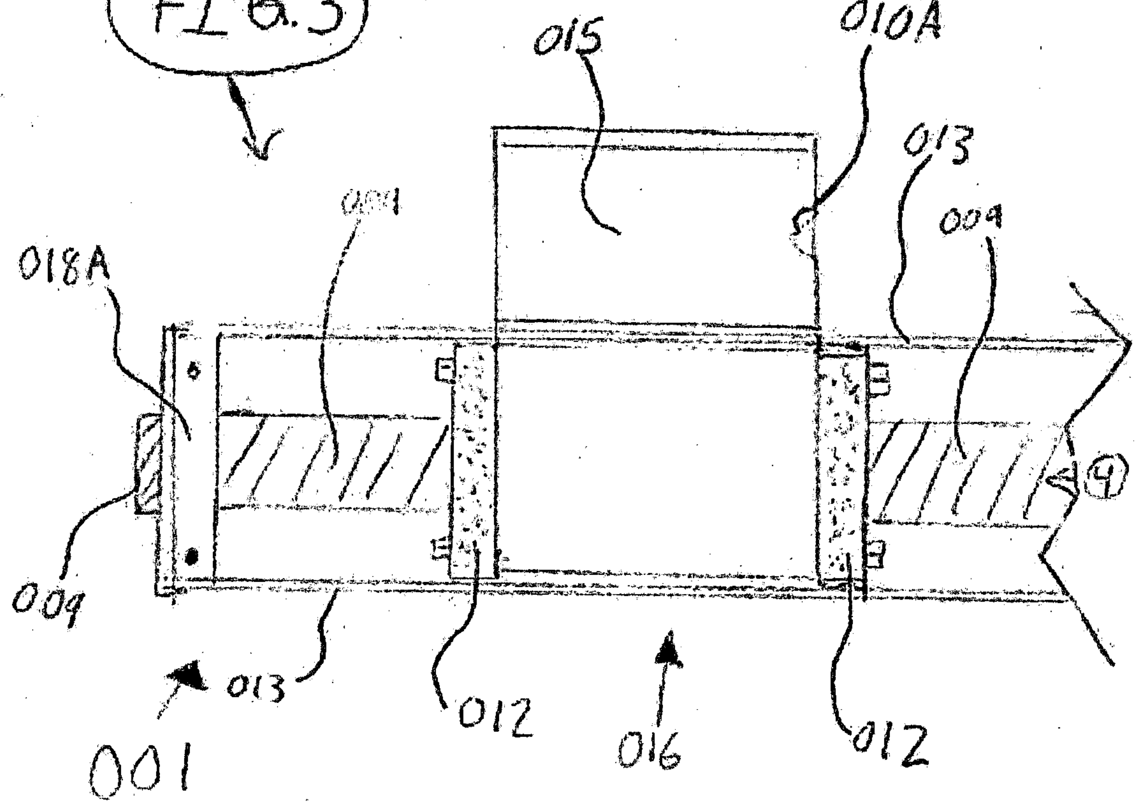


FIG. 3



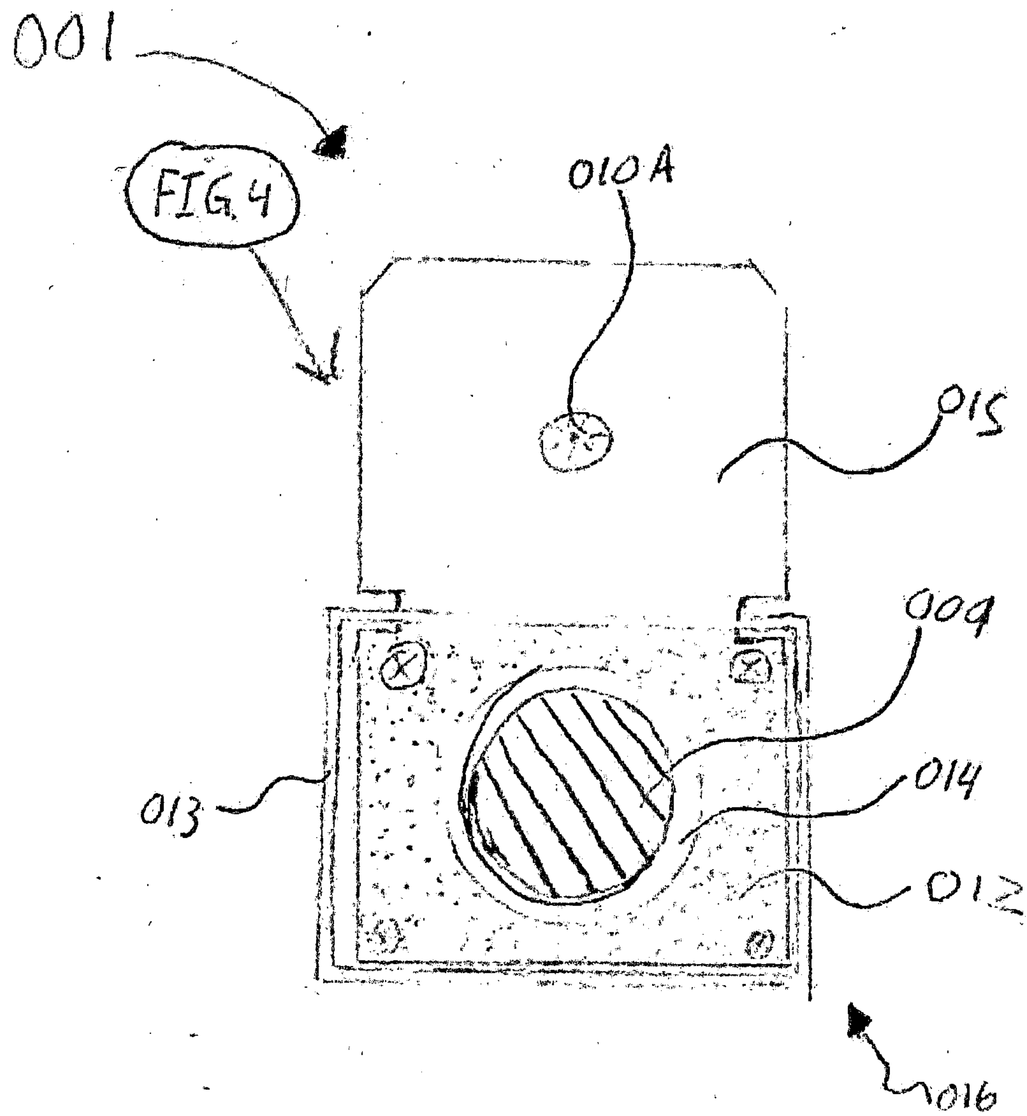
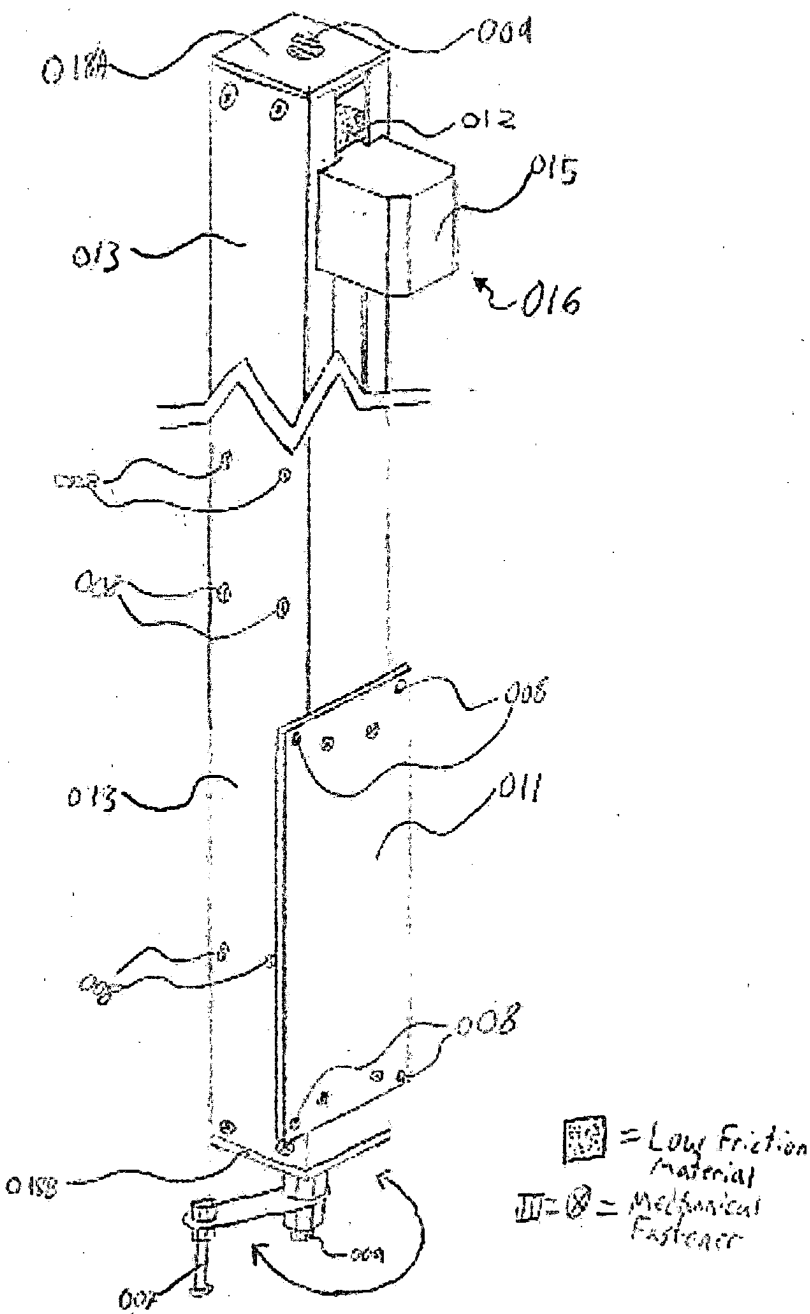

001
FIG. 4
010A
015
009
014
013
012
016

FIG. 5



Drawing NOT to scale

FIG. 6

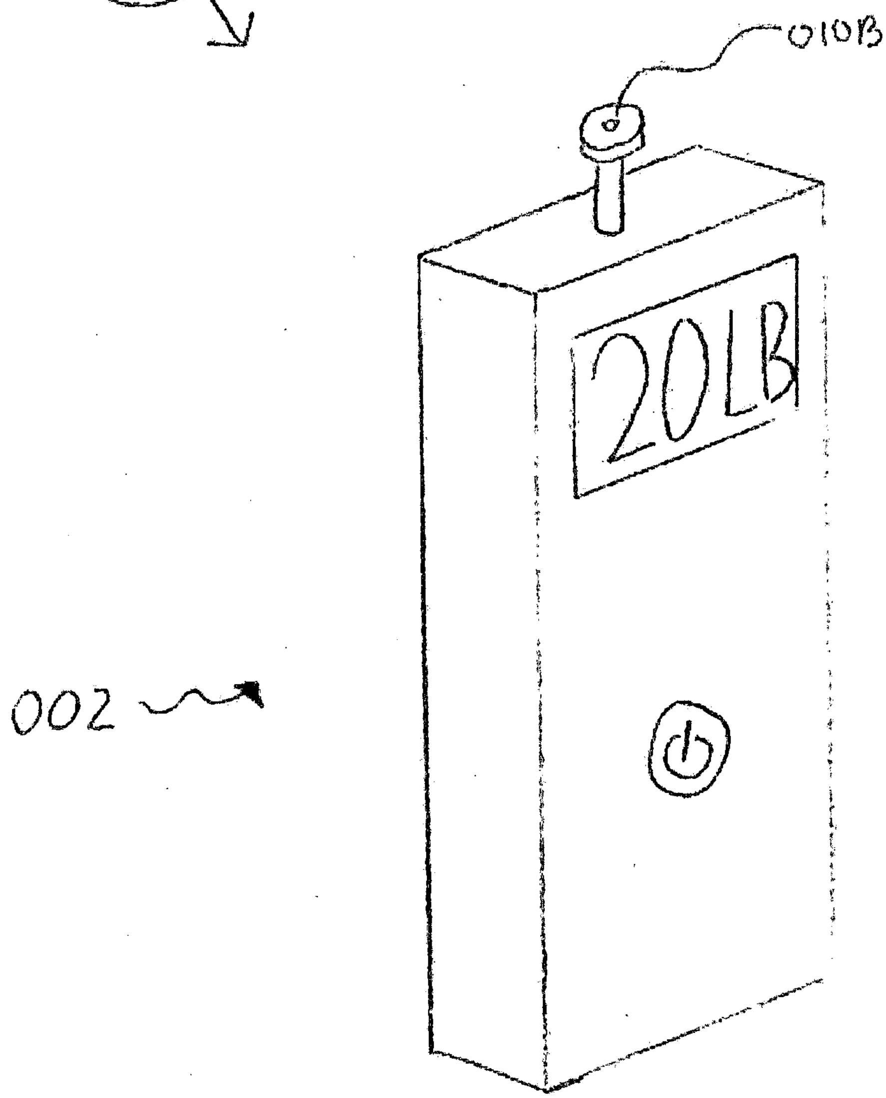
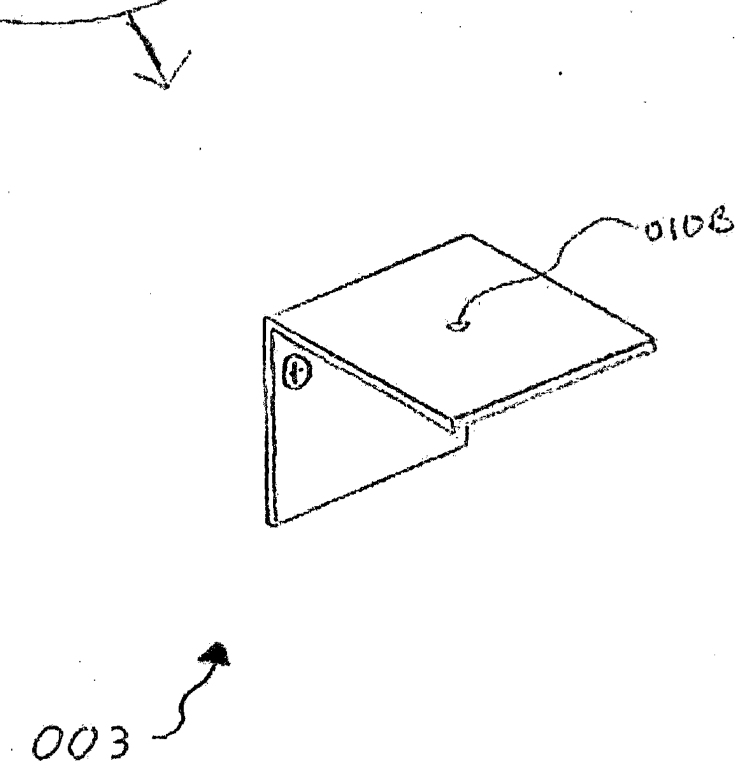


FIG. 7



⊗ = Mechanical Fastener

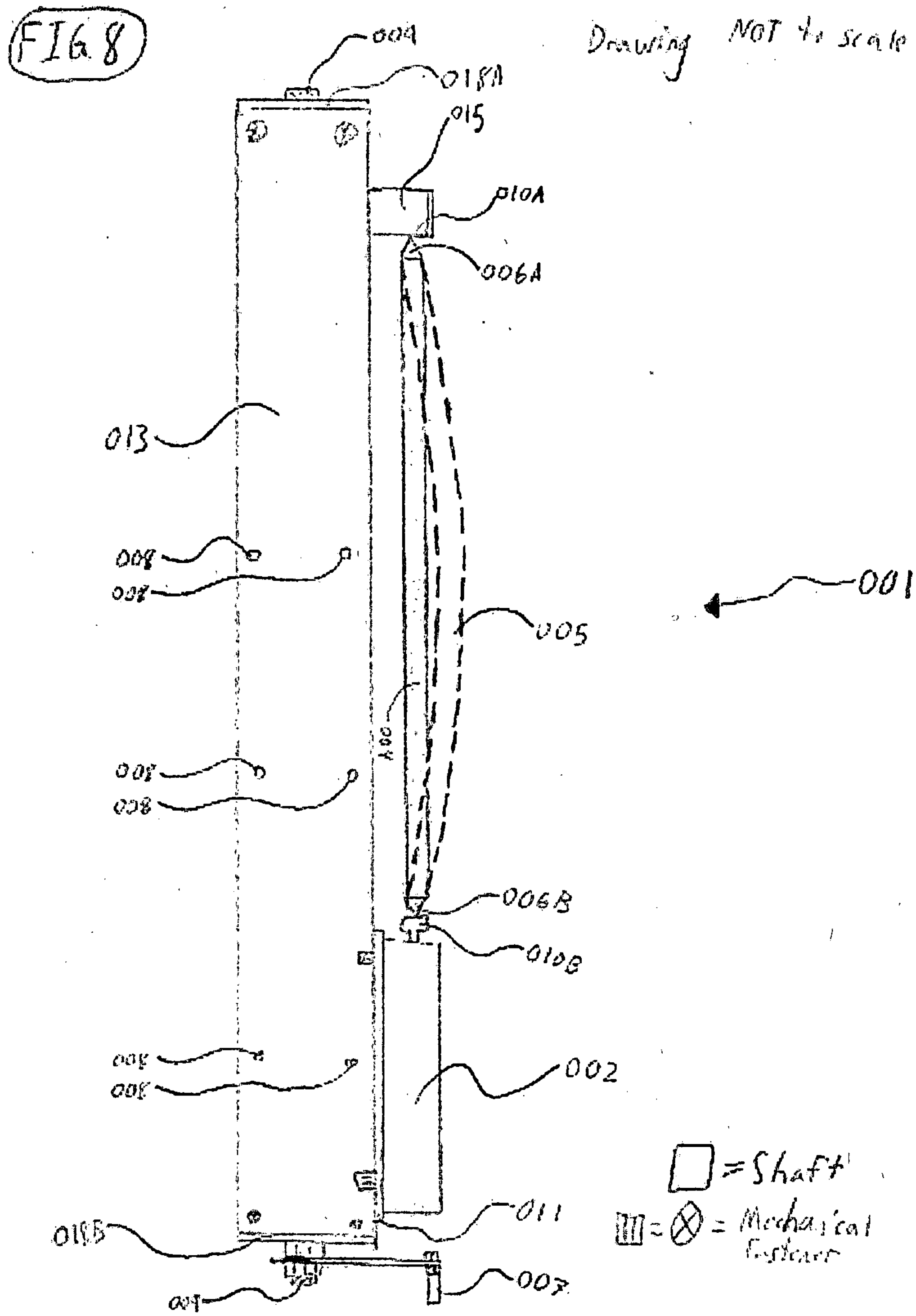
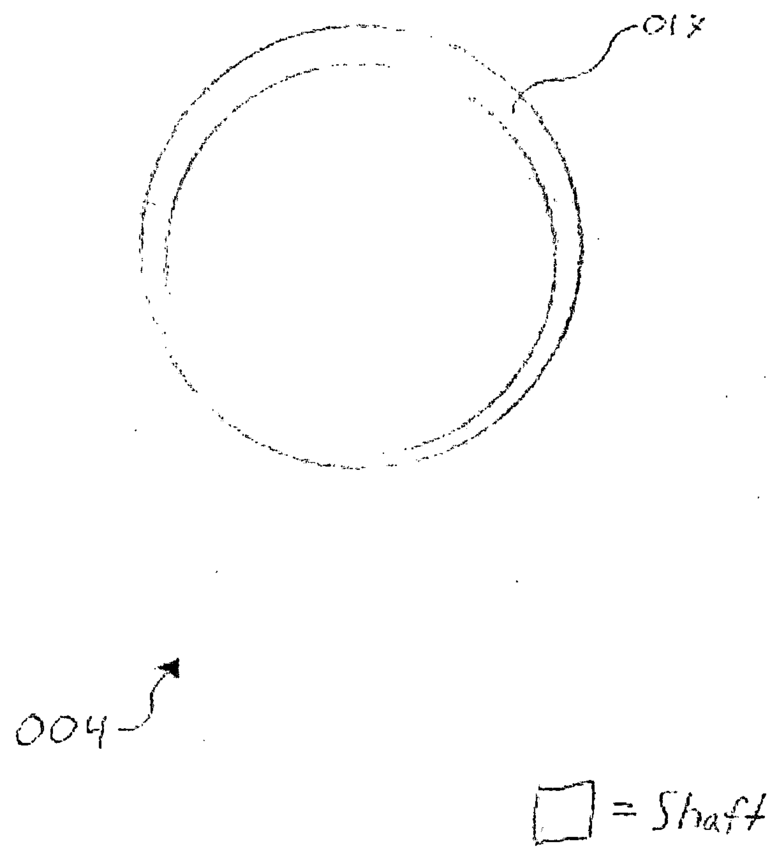


FIG 8
Drawing NOT to scale
004
018A
015
010A
006A
013
008
008
001
005
004
008
008
006B
010B
008
008
002
= Shaft
= ⊗ = Mechanical fastener
011
018B
009
007

FIG. 9

Drawing NOT to scale



METHOD AND APPARATUS FOR INDEXING AND BATCHING SHAFTS

BACKGROUND

[0001] In real world products there is never perfection, some examples could be, multiple archery arrows that each deflect differently under the load of a bow assembly, a golf club shaft wherein the first dynamic bend is not oriented correctly for the swing of the golfer, or structural asymmetry in a shaft causing the existence of a first dynamic bend. However, there are apparatuses and methods to enhance the consistency of those shafts. Individuals typically want their shafts to have as little deviation between each shaft as possible, and to orient the structural asymmetries in the shaft. Most methods and apparatuses for making shafts more consistent are not to the very high standard of what most individuals desire.

SUMMARY

[0002] A variety of additional inventive aspects will be set forth in the description that follows. The inventive aspects can relate to individual features and to combinations of features. It is to be understood that both the forgoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the broad inventive concepts upon which the embodiments disclosed herein are based.

[0003] In general terms, this disclosure is directed to a method for indexing and batching shafts. In some embodiments the method for indexing and batching shafts includes inserting a shaft into an indexing and batching apparatus, operating the apparatus resulting in longitudinal force being applied to the shaft, observing and measuring the properties of the shaft under load, allowing the batching of shafts with similar or identical characteristics.

[0004] In general terms this disclosure is also directed to an apparatus for indexing and batching shafts. In some embodiments the apparatus for indexing and batching shafts includes a body having upper and lower portions as well as inner and outer portions, a mechanism for applying force to a shaft longitudinally, and the apparatus is configured so the applied force causes the shaft to deflect along the first dynamic bend.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The accompanying drawings, which are incorporated in and constitute a part of the description, illustrate several aspects of the present disclosure. A brief description of the drawings is as follows:

[0006] FIG. **1** is a front view with a break line of an apparatus for indexing and batching arrows according to an embodiment of the disclosure.

[0007] FIG. **2** is a side view of the lower portion of the apparatus depicting the lower internal and external mechanisms according to an embodiment of the disclosure.

[0008] FIG. **3** is a side view of the upper portion of the apparatus depicting the upper internal and external mechanisms according to an embodiment of the disclosure.

[0009] FIG. **4** is a bottom view of the upper portion of the apparatus depicting a different view of the upper internal and external mechanisms according to an embodiment of the disclosure.

[0010] FIG. **5** is a perspective view of the apparatus with a break, depicting the upper portion, lower portion, and hand crank according to an embodiment of the disclosure.

[0011] FIG. **6** is a perspective view of a force gauge that could be attached to the apparatus according to an embodiment of the disclosure.

[0012] FIG. **7** is a perspective view of an indexing bracket that could be fastened to the apparatus according to an embodiment of the disclosure.

[0013] FIG. **8** is a side view of the apparatus depicting the apparatus with a shaft and a shaft deflecting along the first dynamic bend according to an embodiment of the disclosure.

[0014] FIG. **9** is a cross section view of an embodiment of a shaft exaggerating the structural asymmetries in the shaft wall that may not be readily apparent in real world situations.

[0015] Each portion of the apparatus is labeled with numerals as follows, indexing and batching apparatus **001**, force gauge **002**, indexing bracket **003**, shaft **004**, shaft deflected along the first dynamic bend **005**, pivot points **006** (A/B), crank handle (rotating) **007**, mounting holes for extra appendages **008**, threaded rod **009**, receiving depressions (A/B) **010**, mounting plate **011**, low friction guide **012**, body **013**, threaded receiver **014**, sliding block **015**, rail system **016**, shaft wall **017**, guide for threaded rod (A/B) **018**.

DETAILED DESCRIPTION

[0016] Various embodiments will be described in detail with reference to the drawings, wherein like reference numerals represent like parts and assemblies throughout the several views. Reference to various embodiments does not limit the scope of the claims attached hereto. Additionally, any examples set forth in this specification are not intended to be limiting and merely set forth some of the many possible embodiments for the appended claims.

[0017] The present disclosure generally provides a method and apparatus to make the properties of shafts more apparent.

[0018] Referring to FIG. **1** an embodiment of an indexing and batching apparatus **001** without the optional appendages to receive the lower end of the shaft is depicted. Examples of shafts **004** may be archery arrow shafts, golf shafts, and spear shafts. FIG. **1** depicts an embodiment of the apparatus with a break line that separates the upper and lower portions of the apparatus.

[0019] Referring to FIG. **1**, an embodiment of the indexing and batching apparatus **001** is depicted with a body **013**, hand crank **007**, mounting plate **011**, sliding block assembly **015**, rail system **016**, threaded rod **009**, mounting holes **008** for extra appendages, upper receiving depression **010**A, and mounting plate **011**. The embodiment pictured depicts a body **013** with threaded rod **009** running through the length of the body **013** secured with mechanical fasteners at the lower end of the body **013** to the crank handle **007**. The upper and lower ends of the threaded rod **009** are able to free spin within upper and lower guides for threaded rod **018**. As depicted in the embodiment in FIG. **2**, attached to the lower end of the threaded rod **009** is a crank handle **007** held in place on the threaded rod **009** by mechanical fasteners. The threaded rod is held in place, but not restricted from rotating, above and below the lower guide for threaded rod **018**B by mechanical fasteners.

[0020] Referring to FIG. **1**, an embodiment of an indexing and batching apparatus, is depicted with a break in the midsection to draw attention to the ends of the apparatus. The upper portion of the apparatus is depicted with a rail system **016**, a sliding block **015**, and threaded rod **009**. As seen in FIG. **3** the body **013** acts as a rail to the sliding block **015** which is held on the rail by a low friction guide **012**. As depicted in FIG. **4** the sliding block **015** may be held in place on the threaded rod **009** by a threaded receiver **014**. As the crank handle **007** is rotated, the threaded rod **009** transfers torque to the threaded receiver **014** which in turn moves the sliding block **015** along the rail system **016**.

[0021] Referring to FIG. **5** and embodiment of an indexing and batching apparatus **001** is depicted with a mounting plate **011** for attaching optional appendages. These appendages may include a gauge **002** (depicted in FIG. **6**) or an indexing bracket **003** (depicted in FIG. **7**). Any commercially available force gauge may be fitted on the mounting plate **011** provided it has the correct mounting structures. Each appendage attached to the mounting plate is configured to receive the lower pivot point **006**B with a receiving depression **010**B.

[0022] Referring to FIG. **8** an embodiment of an indexing and batching apparatus **001** is depicted with a force gauge **002** and a shaft **004** in the vertical position. The shaft **004** is held in place by two pivot points. **006** on both ends of the shaft and are received by receiving depressions **010**. An individual rotates the crank handle **007** that then moves the sliding block **015** along the rail system **016** which applies force to the shaft **004** longitudinally. When the shaft is positioned vertically in space, there is virtually no gravity effect on the deflection of the shaft. When force is applied to the shaft longitudinally, the shaft deflects along the first dynamic bend (as depicted in FIG. **8**). The first dynamic bend of a shaft is a result of real-world structural asymmetries (as depicted in FIG. **9**) that cause a shaft to deflect a certain direction based on the location of these asymmetries. The individual then observes the location of the first dynamic bend. Applying increasing amounts of force to the shaft **004** causes it to deflect (refer to **005**) and there the individual is able to measure the longitudinal force it takes to deflect the shaft **004** a measurable amount which correlates to the resonance frequency of the shaft and the elasticity of the shaft. Once the first dynamic bend and other properties are determined, the individual can then index and batch shafts **004** to those properties.

What is claimed is:

1. A method for indexing and batching shafts, comprising;
inserting a shaft into an indexing and batching apparatus;
operating the apparatus resulting in longitudinal force being applied to the shaft;
observing and measuring the properties of the shaft under load;
batching each shaft based on its specific properties.

2. The method of claim **1**, wherein the body and shaft may be positioned vertically.

3. The method of claim **1**, wherein the shaft is flexed on a plurality of concentric pivot points.

4. The method of claim **1**, wherein the shaft is flexing along the first dynamic bend.

5. The method of claim **4**, wherein these properties are observed and measured;
the location of the first dynamic bend on the shaft
the amount of deflection of a shaft while flexing along the first dynamic bend, under load;
the amount of force applied to the shaft to result in a measurable deflection;

6. An apparatus for indexing and batching shafts comprising;
a body having upper and lower portions as well as inner and outer portions;
a mechanism for applying force to a shaft longitudinally, but not limited to;
the applied force causes the shaft to deflect along the first dynamic bend.

7. The apparatus of claim **6**, wherein the body of the apparatus is sized to accommodate the deflection of the shaft.

8. The apparatus of claim **6**, wherein the mechanism is applying force to locate to first dynamic bend.

9. The apparatus of claim **6**, wherein the body is sized to accommodate varying sizes of shafts.

10. The apparatus of claim **6**, wherein the indexing and batching apparatus may be configured to be operated manually.

11. The apparatus of claim **6**, wherein the indexing and batching apparatus may be configured to receive optional appendages.

12. The apparatus of claim **6**, wherein the indexing and batching apparatus may contain a threaded system that allows a sliding block assembly to apply force to a shaft.

13. The apparatus of claim **12**, wherein the apparatus contains a hand crank that rotates and mechanically fastens a threaded rod that then moves the sliding block assembly to apply force to a shaft.

14. The apparatus of claim **12**, wherein the sliding block assembly is encased in a rail system.

15. The apparatus of claim **12**, wherein the sliding block assembly attaches to and is moved by the threaded rod along a rail system on and/or in the body of the apparatus;
the sliding block comprising;
a portion that attaches the block to the threaded rod;
a low friction material that contacts the rail system; and
a receiver for an end of a shaft

16. The apparatus of claim **12**, wherein attached to the body is a mounting plate for optional appendages, such as but not limited to, a gauge.

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